Predicting Race Outcome Deviations in Formula 1

A PROJECT REPORT

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Writer Identification from Text Blogs using Statistical and Similarity-Based Feature Analysis

Abstract —

The results of Formula One races depend on various interrelated elements including initial standings in qualifying sessions, overall team efficiency, individual drivers' abilities, and unique occurrences during each event. Current methods for forecasting tend to concentrate on identifying likely champions based largely on chance factors and unforeseen events. The current endeavor concentrates on identifying discrepancies in drivers' starting positions relative to their finishing places during races. To forecast how well a racer performs relative to their usual standards in an event. Machine learning algorithms under supervision classify deviations, whereas those without it examine discrepancies among model outputs. A thorough examination of errors reveals insights into prediction variability and constraints imposed by models. The endeavor showcases how various techniques such as data preparation, machine learning algorithms like classification and clustering, along with evaluating errors, align with what is expected in this academic program's objectives.

Keywords —

1. Formula 1 Analytics
2. Performance Deviation
3. Supervised Learning
4. Unsupervised Learning
5. Error Analysis

I. Introduction

In Formula One racing, decision-making relies heavily on extensive data analysis. Driver skill, vehicle performance, race strategy, and unpredictable factors such as weather conditions and accidents significantly influence race outcomes. Traditional predictive approaches primarily focus on forecasting race winners, which often leads to unstable results due to inherent uncertainties in racing events.

This project introduces a deviation-based analytical approach, where the focus shifts from predicting winners to analyzing the difference between qualifying positions and final race positions. This deviation-based modeling provides a more stable and interpretable representation of performance.

In addition to previously implemented distance-based methods such as k-Nearest Neighbors (k-NN), this phase of the project incorporates **Decision Tree learning techniques**. Decision Trees use impurity measures such as **Entropy and Gini Index** to split the dataset and identify the most informative features. This allows the model to capture hierarchical decision rules that explain performance deviations. Furthermore, continuous numerical attributes are transformed into categorical values using **binning techniques**, enabling the effective application of tree-based algorithms. The study also includes visualization of decision trees and decision boundaries, providing intuitive insights into how different features influence classification.

This combined approach enhances interpretability, improves understanding of feature importance, and provides deeper insights into performance variability in Formula One racing.

II. Literature Review

Earlier research into athletic data science highlights how machine learning enhances predictive capabilities and analytical insights within sporting contexts. Studies indicate that factors like past performances of teams and their trustworthiness play crucial roles in determining racing results. Nevertheless, literature underscores how predictions about race winners can be greatly influenced by unforeseen circumstances.

Research focusing on model classifications highlights that converting numerical results into discrete categories typically enhances understanding and stability of predictions. Studies indicate that when multiple models disagree significantly, it frequently signifies uncertain or imprecise data elements.

Techniques like unsupervised learning's clustering method is commonly employed to uncover hidden structures within intricate data sets. Research in error analysis indicates that examining mistakes offers more comprehensive understanding of systems' functioning compared to relying solely on performance indicators. This research inspires employing agreement-focused evaluation techniques within our study.

Existing literature on k-NN classifiers highlights that their effectiveness is strongly influenced by **data distribution and class overlap**. Studies emphasize that smaller neighborhood sizes tend to overfit noisy data, while larger neighborhood sizes may oversmooth meaningful patterns. Prior research also suggests that performance metrics alone are insufficient for evaluation, and that comparing training and testing behavior is essential to identify overfitting and underfitting. These insights directly motivate the experimental analysis conducted in this project.

Recent studies in machine learning emphasize the importance of tree-based models such as Decision Trees for classification problems due to their interpretability and ability to handle both categorical and continuous data. Research highlights that impurity measures like entropy and Gini index are effective in selecting optimal splitting features, thereby improving classification accuracy. Additionally, discretization techniques such as equal-width and equal-frequency binning are widely used to convert continuous data into categorical form, making them suitable for algorithms that rely on categorical inputs. Visualization of decision trees and decision boundaries has also been recognized as an essential tool for understanding model behavior and validating decision-making processes. These insights form the foundation for incorporating Decision Tree-based analysis in this project.

III. Methodology

A. System Overview

The objective of this phase of the Formula One performance deviation project is to analyze similarity and classification behavior using statistical and distance-based techniques. Instead of directly predicting race winners, this project focuses on identifying whether a driver **outperforms, underperforms, or performs as expected** relative to their qualifying position.
Lab-3 concepts were applied to historical Formula One race data by converting race records into numerical feature vectors, analyzing statistical properties of driver performance, computing similarity using distance metrics, and applying k-Nearest Neighbors (k-NN) classification to categorize race outcomes.

B. Flow Diagram

The flow diagram represents the overall processing pipeline followed in the proposed system.

Input F1 ace & Qualifying Data
↓
Data Cleaning & Preprocessing
↓
Feature Engineering
↓
Deviation Label Generation
↓
Supervised Classification
↓
Model Disagreement Detection
↓
Unsupervised Clustering
↓
Performance Interpretation

C. Architecture Description

Input Module:
Collects ancient race consequences, qualifying positions, constructor details, circuit data, and race popularity from a public components 1 information source.
Preprocessing Module:
Handles lacking values, filters DNFs wherein required, encodes specific variables, and scales numerical functions to make sure truthful assessment.
characteristic Engineering Module:
Computes race outcome deviation and ancient facts along with common end role, rolling deviation, DNF rate, and constructor reliability.

Deviation Labeling Module:
Converts deviation values into 3 instructions: Outperform, neutral, and Underperform.

Supervised Module:
Trains multiple category fashions including Logistic Regression, help Vector device, and Random woodland to predict deviation classes.

Confrontation analysis Module:
Identifies race instances wherein special fashions expect distinct deviation classes, indicating uncertainty.
Unsupervised Module:
Clusters disagreement samples to find out hidden styles together with risky circuits or method-touchy races.
choice Module:
Produces final analytical insights and model overall performance summaries.

D. Parameter Selection and Justification

Parameter	**Value**	**Justification**
Deviation Threshold	+2 positions	Reduces noise from minor position changes
Number of Classes	3	Outperform, Neutral, Underperform
Classification Models	LR, SVM, RF	Covers linear and non-linear behavior
Clustering Algorithm	K-Means	Simple and interpretable
Number of Clusters	3–5	Balances detail and clarity

E. Feature Representation and Vector Operations (F1)

In the F1 dataset, each race entry was treated as a vector containing features such as qualifying grid position, average finishing position, constructor reliability, and deviation from expected performance. Dot product and Euclidean norm were implemented to understand similarity and magnitude of driver performance vectors. These operations helped in understanding how closely two race performances or two drivers resemble each other in multi-dimensional feature space. The correctness of the implementations was verified using NumPy functions.

F. Minkowski Distance Analysis (F1)

Minkowski distance was computed between selected F1 race vectors for values of p ranging from 1 to 10. The variation in distance values demonstrated how sensitivity to feature differences increases with higher values of p.
This experiment highlighted how distance metric selection directly affects neighbor selection in k-NN classification of F1 race outcomes.

G. k-NN Classification (F1)

The F1 dataset was divided into training and testing sets using a two-class deviation setup. A k-NN classifier with $k = 3$ was trained to classify races into outperform or underperform categories.
Prediction accuracy and confusion matrices were analyzed to evaluate classification effectiveness.

H. Decision Tree Learning and Impurity Measures

In this phase, Decision Tree learning was implemented to classify performance deviation in Formula One data.
Entropy Calculation:
Entropy measures the randomness or impurity in the dataset. Higher entropy indicates more disorder, while lower entropy indicates more homogeneous data.
Gini Index:
The Gini Index is another impurity measure that evaluates the probability of incorrect classification of a randomly chosen element.
Information Gain:
Information Gain is used to select the best feature for splitting the dataset. It measures the reduction in entropy after a dataset is split based on a feature.
Binning Techniques:
Since many features in the dataset are continuous, equal-width and equal-frequency binning methods were applied to convert them into categorical values.
Decision Tree Construction:
A custom Decision Tree was built using recursive partitioning based on Information Gain. The root node represents the feature with the highest information gain, and subsequent splits form a hierarchical decision structure.
Visualization:
The constructed Decision Tree was visualized using standard plotting techniques to interpret the decision-making process.
Decision Boundary Analysis:
Using two selected features, the decision boundary of the Decision Tree classifier was plotted to understand how the feature space is partitioned.

IV. Results Analysis & Discussion

A. Intraclass Spread and Interclass Distance in F1 Performance Deviation

Intraclass spread and interclass distance were analyzed to evaluate how well the deviation-based performance classes in the

Formula One dataset were separated. The classes considered were **Outperform** and **Underperform**, derived from the difference between qualifying position and final race position.

The intraclass spread represented the variability in driver performance within the same deviation class. Results showed that drivers classified as outperforming exhibited relatively higher spread, indicating that exceptional performances occur under varying race conditions. In contrast, underperforming instances demonstrated comparatively lower spread, suggesting consistency in negative deviation patterns.

B. Feature Distribution and Density Observations

Histogram analysis of selected F1 features, such as qualifying position and deviation magnitude, revealed non-uniform distribution patterns. Mid-grid qualifying positions showed higher variance in deviation, reflecting greater uncertainty in race outcomes due to overtaking opportunities and strategic variations.

Front-grid positions demonstrated lower variance, indicating more stable performance, while back-grid positions showed limited positive deviation potential. These distribution patterns influenced distance calculations and neighbor selection during classification.

C. Behavior of Distance Metrics on F1 Race Data

The Minkowski distance metric was evaluated for values of p ranging from 1 to 10 using selected F1 race vectors. Lower values of p resulted in smoother distance variation, while higher values amplified feature differences.
It was observed that excessively large values of p increased sensitivity to minor feature variations, which negatively affected neighbor stability. Moderate values of p provided a balanced representation of performance similarity, making them more suitable for F1 race classification tasks.

D. k-Nearest Neighbors Classification Results

A k-Nearest Neighbors classifier with $k = 3$ was applied to classify F1 race instances into outperform and underperform categories. The classifier achieved consistent accuracy across both training and testing datasets, confirming its ability to generalize performance patterns rather than memorizing individual race instances. Prediction results showed that most misclassifications occurred near class boundaries, particularly for drivers whose finishing position closely matched their qualifying position. This highlights the inherent uncertainty present in tightly contested races.

E. Effect of k on Classification Performance

The value of k was varied from 1 to 11 to study its impact on classification accuracy. When $k = 1$, the classifier exhibited high sensitivity to individual race outcomes, leading to overfitting. This resulted in higher training accuracy but lower testing accuracy.

As k increased, the decision boundary became smoother, reducing sensitivity to noisy race events. However, very large values of k caused underfitting by oversmoothing meaningful deviation patterns. An intermediate value of k achieved balanced performance, indicating an optimal trade-off between bias and variance.

F. Model Fit Interpretation: Overfitting and Underfitting

Model fit was analyzed by comparing training and testing metrics across different values of *k*. Overfitting was observed at lower values of *k* due to excessive reliance on individual race instances. Underfitting occurred at higher values of *k*, where meaningful deviation trends were lost.
For moderate values of *k*, the classifier demonstrated stable performance, indicating a regular fit. This balanced behavior is desirable for real-world F1 performance analysis, where robustness to noise and variability is essential.

G. Class Separability and Distance-Based Behavior

Based on intraclass spread and interclass distance observations, the deviation-based classes in the F1 dataset are reasonably but not perfectly separated. Extreme outperform and underperform cases form distinct regions in feature space, while instances close to neutral deviation introduce overlap. This overlap explains misclassifications near class boundaries and reflects the inherent uncertainty present in closely contested races.

The behavior of distance metrics further influences classification. As neighborhood size increases, sensitivity to individual race instances decreases, resulting in smoother decision boundaries. However, excessive smoothing reduces the model's ability to capture meaningful local deviation patterns.

H. Effect of k on k-NN Performance

The performance of the k-NN classifier varies significantly with the choice of k. Smaller values of k lead to high sensitivity to individual race outcomes, capturing noise along with meaningful patterns. This results in higher training accuracy but reduced testing performance. As k increases, the classifier becomes more stable by reducing the influence of outliers.

However, very large values of k dilute local deviation information, leading to reduced classification accuracy. An intermediate value of k provides a balanced representation of neighborhood influence, achieving improved generalization performance.

I. Model Fit, Overfitting, and Suitability of k-NN

Overfitting is observed when k is small, as predictions rely excessively on nearby individual samples. Underfitting occurs at higher values of k where meaningful deviation trends are averaged out. For moderate values of k, training and testing performances are closely aligned, indicating a regular fit.
Based on these observations, k-NN is a suitable classifier for deviation-based F1 performance analysis. When applied with appropriate feature scaling and parameter selection, it effectively captures performance patterns while maintaining generalization capability.

J. Decision Tree Analysis Results

The Decision Tree model was applied to classify deviation categories in the dataset. The entropy and Gini index values provided insights into the impurity of the dataset before splitting.
The feature selected as the root node using Information Gain was found to be the most influential attribute affecting race outcome deviation. This demonstrates the effectiveness of Information Gain in identifying dominant features.
The constructed Decision Tree showed a clear hierarchical structure, where early

splits significantly reduced uncertainty. Visualization of the tree provided interpretability by showing how decisions are made at each node.
Decision boundary analysis using two selected features revealed that the Decision Tree creates non-linear partitioning of the feature space. This allows the model to capture complex relationships between features and deviation classes.

K. Observations and Inferences

Entropy vs Gini:
Both impurity measures provided similar trends, but Gini Index was computationally simpler and faster.

Effect of Binning:
Converting continuous features into categorical values improved Decision Tree performance and interpretability.

Feature Importance:
Information Gain effectively identified the most relevant feature, which significantly influenced classification.

Decision Tree Behavior:
The model created hierarchical rules that were easy to interpret compared to distance-based models.

Decision Boundary:
The Decision Tree produced non-linear decision boundaries, enabling better separation of complex patterns.

Model Comparison Insight:
Compared to k-NN, Decision Trees provide better interpretability but may suffer from overfitting if not controlled.

XVI. Conclusion

This project successfully explored deviation-based analysis of Formula One race outcomes using multiple machine learning techniques. While earlier phases focused on distance-based classification using k-NN, this phase introduced Decision Tree learning to enhance interpretability and feature-based analysis. The use of entropy, Gini index, and Information Gain enabled effective identification of important features influencing race performance. Binning techniques allowed continuous data to be transformed into categorical values, facilitating better model performance. Decision Tree visualization and decision boundary analysis provided deeper insights into how classification decisions are made. The results highlight the importance of selecting appropriate models based on problem requirements, balancing interpretability and performance.
Overall, this work demonstrates how combining statistical methods, distance-based learning, and tree-based models can provide a comprehensive understanding of complex real-world datasets such as Formula One race analytics.

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